

SERAK, Czech Republic

WMO: 117300

Lat: 50.188N

Lon: 17.108E

Elev: 1327

StdP: 86.36

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-13.9	-23.4	0.5	-8.9	-19.7	0.8	-7.9	18.9	-4.9	17.4	-4.7	7.5	100	0.816

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	22.4	15.4	20.6	14.9	19.1	14.3	16.5	20.2	15.7	19.3	14.9	18.4	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	12.7	17.9	14.2	11.9	17.3	13.3	11.2	16.6	51.4	20.4	48.7	19.2	46.3	18.5	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.5	12.1	-18.7	24.9	4.0	1.3	-21.6	25.9	-24.0	26.7	-26.2	27.4	-29.1	28.4		
(5)			-19.0	18.3	3.9	0.8	-21.9	18.9	-24.2	19.4	-26.4	19.9	-29.2	20.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-5.4	-5.3	-2.4	3.4	7.2	11.1	13.1	13.1	8.6	4.3	0.2	-3.5		
	DBStd	7.94	4.18	4.78	4.76	4.66	4.43	4.20	3.86	3.73	4.00	4.39	4.57	4.13		
	HDD10.0	2631	477	428	384	201	109	38	14	12	74	181	293	420		
	HDD18.3	5333	736	662	643	447	344	220	166	167	294	434	543	678		
	CDD10.0	352	0	0	0	4	23	69	111	108	31	5	0	0		
	CDD18.3	12	0	0	0	0	0	2	5	5	0	0	0	0		
	CDH23.3	18	0	0	0	0	0	1	9	7	1	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.0	7.6	7.0	6.5	5.4	5.1	4.6	4.5	4.5	5.4	6.4	7.2	7.6		
(15) Precipitation	PrecAvg	879	60	53	64	55	83	101	119	86	79	59	60	60		
	PrecMax	1214	128	112	151	90	216	183	506	199	180	164	99	114		
	PrecMin	595	25	12	23	6	34	37	32	23	18	8	0	17		
	PrecStd	125	28	25	29	22	38	36	90	41	45	36	23	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	6.7	9.5	16.3	20.2	23.0	24.6	24.2	19.2	16.2	12.2	6.6		
		MCWB	1.3	-0.1	3.9	9.7	13.4	16.4	16.4	15.4	14.0	9.2	4.3	0.8		
	2%	DB	2.8	3.6	7.2	13.5	17.2	20.6	22.5	22.2	17.1	13.1	9.6	4.8		
		MCWB	1.6	0.0	3.6	8.1	12.2	15.0	15.4	15.1	13.0	9.1	5.7	1.1		
	5%	DB	1.5	1.5	5.5	11.6	15.4	18.8	20.6	20.4	15.6	11.4	7.6	3.5		
		MCWB	0.3	0.2	2.5	7.3	11.1	14.3	15.0	14.9	11.9	8.7	5.6	1.2		
	10%	DB	0.1	0.1	3.8	9.7	13.6	17.1	18.9	18.8	13.9	10.0	6.2	1.8		
		MCWB	-0.7	-0.8	2.3	6.2	10.1	13.2	14.2	14.3	11.0	8.1	4.7	0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	2.5	5.6	10.4	14.6	17.5	18.2	17.1	14.9	11.8	8.4	4.1		
		MCDB	3.6	3.3	7.6	14.7	18.8	21.0	21.8	21.0	18.2	13.1	9.3	4.4		
	2%	WB	2.0	1.4	4.4	8.8	12.9	16.2	16.5	16.2	13.6	10.2	7.2	3.1		
		MCDB	2.3	2.0	6.1	12.6	16.3	19.2	20.5	19.9	16.3	11.6	8.0	3.7		
	5%	WB	0.7	0.3	3.4	7.7	11.7	14.9	15.6	15.5	12.5	9.2	6.1	1.9		
		MCDB	1.2	1.1	4.9	10.9	14.5	18.2	19.4	19.1	14.6	10.6	7.1	2.6		
	10%	WB	-0.5	-0.7	2.3	6.6	10.5	13.7	14.7	14.7	11.5	8.3	5.0	0.5		
		MCDB	-0.2	0.3	3.8	9.3	13.2	16.5	18.2	18.4	13.4	9.7	5.8	1.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.1	4.7	5.7	6.0	6.0	6.2	6.2	5.0	4.5	4.1	3.9		
		MCDBR	5.0	5.7	6.4	7.4	7.6	7.7	8.0	7.7	6.5	5.8	5.7	5.4		
	5% WB	MCWBR	4.2	4.4	4.4	4.3	4.5	4.8	4.5	4.2	4.4	4.1	3.9	4.0		
		MCDWR	4.5	5.1	5.7	6.9	7.2	7.3	7.4	7.1	6.1	5.1	4.7	4.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.262	0.283	0.315	0.354	0.352	0.352	0.364	0.358	0.332	0.312	0.291	0.261		
		taud	2.392	2.391	2.377	2.327	2.354	2.371	2.334	2.364	2.428	2.471	2.476	2.440		
	Ebn at Noon	Ebn at Noon	797	866	894	889	904	905	886	874	864	820	755	749		
		Edh at Noon	61	80	97	115	117	116	119	110	92	74	58	51		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.84	1.58	2.64	4.14	5.01	5.31	5.05	4.54	3.21	1.87	0.92	0.63		
	RadStd	RadStd	0.15	0.24	0.43	0.52	0.60	0.50	0.57	0.42	0.47	0.28	0.12	0.07		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (45 neighbors)		+0.50	N/A	N/A	+0.75	+0.35	+0.31	N/A	-152	+65	+27	

Nomenclature: See separate page